

(†) The ring R is a commutative ring with identity.

Let V and W be two free R -modules of finite rank with basis $\{v_1, v_2, \dots, v_n\}$ and $\{w_1, w_2, \dots, w_m\}$ respectively

Lemma 0.1. : For $1 \leq k \leq n$ and $1 \leq l \leq m$, there exists R -module homomorphisms $f^{kl} : V \otimes W \rightarrow R$ such that

$$f^{kl}(v_i \otimes w_j) = \delta_{(i,j)}^{(k,l)}$$

for $\forall 1 \leq i \leq n$ and $1 \leq j \leq m$.

Proof: Define

$$f : V \times W \rightarrow R$$

$$(v_i, w_j) \mapsto \delta_{(i,j)}^{(k,l)}$$

then it's clear that f is R -bilinear. By the universal mapping property of tensor product, there exists a unique R -module homomorphism

$$f^{kl} : V \otimes W \rightarrow R$$

such that $f^{kl} \circ \otimes = f^{kl}$, i.e., $f^{kl}(v_i \otimes w_j) = \delta_{(i,j)}^{(k,l)}$. $\square$

Theorem 0.1. : Let V and W be two free R -modules of finite rank with basis $\{v_1, v_2, \dots, v_n\}$ and $\{w_1, w_2, \dots, w_m\}$ respectively, then

$$\{v_i \otimes w_j | 1 \leq i \leq n, 1 \leq j \leq m\}$$

is a basis for $V \otimes W$.

Proof: For $v \triangleq \sum_{i=1}^n a^i v_i \in V$ and $w \triangleq \sum_{j=1}^m b^j w_j \in W$, then $v \otimes w = \sum_{i,j} a^i b^j v_i \otimes w_j$, which means that every element of $V \otimes W$ is in $\text{Span}\{v_i \otimes w_j\}_{i,j}$, $V \otimes W \subseteq \text{Span}\{v_i \otimes w_j\}_{i,j}$. Thus we have $V \otimes W = \text{Span}\{v_i \otimes w_j\}_{i,j}$;

Suppose $\sum_{i,j} c^{ij} v_i \otimes w_j = 0$ with $c^{ij} \in R$. Apply Lemma 0.1, then

$$0 = f^{kl}(\sum_{i,j} c^{ij} v_i \otimes w_j) = \sum_{i,j} c^{ij} f^{kl}(v_i \otimes w_j) = \sum_{i,j} c^{ij} \delta_{(i,j)}^{(k,l)} = c^{kl},$$

which shows that $\{v_i \otimes w_j\}_{i,j}$ is linearly independent. $\square$

By this theorem, we can know that

Corollary 0.1. : If V and W are two free R -modules of finite rank, then

$$\text{rk}(V \otimes W) = \text{rk}(V) \cdot \text{rk}(W)$$

Recall that for the R -modules V and W , $\text{Hom}_R(V, W)$ is an R -module as well w.r.t. the usual addition and scalar multiplication of homomorphisms.

Here, we denote: $V^\vee := \text{Hom}_R(V, R)$.

Proposition 0.1. : Suppose V is a free R -module of rank n with basis $\{e_1, e_2, \dots, e_n\}$. Define

$$\alpha^i : V \rightarrow R$$

$$e_j \mapsto \delta_j^i$$

then $V^\vee$ is a free R -module of rank n with basis $\{\alpha^1, \alpha^2, \dots, \alpha^n\}$.

Proof: For $\forall v \triangleq \sum_{i=1}^n a^i v_i \in V$ and $\forall f \in V^\vee$, we have

$$f(v) = \sum_{i=1}^n a^i f(v_i)$$

note that $\alpha^i(v) = \alpha^i(\sum_{j=1}^n a^j v_j) = \sum_{j=1}^n a^j \alpha^i(v_j) = \sum_{j=1}^n a^j \delta_j^i = a^i$, hence

$$f(v) = \sum_{i=1}^n f(v_i) \alpha^i(v) \quad (\forall v \in V),$$

which means that $f = \sum_{i=1}^n f(v_i) \alpha^i$ and $f \in \text{Span}\{\alpha^1, \alpha^2, \dots, \alpha^n\}$. Thus we have $V^\vee = \text{Span}\{\alpha^1, \alpha^2, \dots, \alpha^n\}$;

Suppose $\sum_{i=1}^n c^i \alpha^i = 0 \in V^\vee$. Apply it to e_j , we get

$$0 = \sum_{i=1}^n c^i \alpha^i(e_j) = \sum_{i=1}^n c^i \delta_j^i = c_j,$$

which shows that $\{\alpha^1, \alpha^2, \dots, \alpha^n\}$ is linearly independent. $\square$

By the proposition above and Lemma 0.1, we can know

Corollary 0.2. : If V and W are free R -modules of rank n and m , with basis $\{v_i\}$ and $\{w_j\}$ respectively, then the linear functions

$$f^{kl} : V \otimes W \rightarrow R$$

$$v_i \otimes w_j \mapsto \delta_{(i,j)}^{(k,l)}$$

forms the basis for $(V \otimes W)^\vee$ dual to the basis $\{v_i \otimes w_j\}$ of $V \otimes W$.

For an R -module V , define

$$T^0(V) := R, \quad T^1(V) := V, \quad T^2(V) := V \otimes V, \dots,$$

$$T^k(V) := V \otimes V \otimes \dots \otimes V \text{ (k times)}$$

and

$$T(V) := \bigoplus_{k=0}^{\infty} T^k(V),$$

here the direct sum $\oplus$ means that each element $v \in T(V)$ is uniquely a finite sum of the form $v = \sum_i v_{k_i}$ with $v_{k_i} \in T^{k_i}(V)$. Then there is a multiplication on $T(V)$: define

$$\mu : T^k(V) \times T^l(V) \rightarrow T^{k+l}(V)$$

$$(x, y) \mapsto x \otimes y,$$

and then extend μ to $T(V) \times T(V) \rightarrow T(V)$ by R -linearity. Then $T(V)$ is a graded R -algebra, called the **tensor algebra** of V .

Now, let $I(V) := \langle v \otimes v \mid v \in V \rangle$ be the ideal generated by all elements of the form $v \otimes v$ in the tensor algebra $T(V)$. Then for $\forall a, b \in T(V)$, $a \otimes v \otimes v \otimes b \in I(V)$.

Exterior Algebra

Definition 0.1. : The **exterior algebra** $\bigwedge(V)$ is defined to be the quotient algebra

$$\bigwedge(V) := T(V)/I(V).$$

We denote the image of $v_1 \otimes v_2 \otimes \cdots \otimes v_k$ by

$$v_1 \wedge v_2 \wedge \cdots \wedge v_k,$$

i.e., $\overline{v_1 \otimes v_2 \otimes \cdots \otimes v_k} = v_1 \wedge v_2 \wedge \cdots \wedge v_k$ (such an element is said to be **decomposable**).

The operation $\wedge$ is called the **wedge product**.

kth Exterior Power

Definition 0.2. : Define the **kth exterior power** $\bigwedge^k(V)$ to be the image of $T^k(V)$ under the projection $T(V) \rightarrow \bigwedge(V) = T(V)/I(V)$. i.e., there is a canonical isomorphism

$$\bigwedge^k(V) \cong \frac{T^k(V)}{T^k(V) \cap I(V)} = \frac{T^k(V)}{I^k(V)},$$

here $I^k(V) = T^k(V) \cap I(V)$ consists of the homogenous elements of degree k in $I(V)$.

Usually, we write $\bigwedge^k V$ for $\bigwedge^k(V)$.

Remark 0.1. : Since $U \otimes (V \otimes W) \cong (U \otimes V) \otimes W \cong U \otimes V \otimes W$, we can know that the wedge product $\wedge$ is associative.

We can derive some properties of wedge product on the exterior algebra

Proposition 0.2. : (1) Let V be an R -module. In the exterior algebra $\bigwedge(V)$, we have

$$u \wedge v = -v \wedge u$$

for all $u, v \in \bigwedge(V)$.

(2) If $u \in \bigwedge^k V$ and $v \in \bigwedge^l V$, then $u \wedge v \in \bigwedge^{k+l} V$ and

$$u \wedge v = (-1)^{kl} v \wedge u.$$

(3) In a decomposable element $v_1 \wedge v_2 \wedge \cdots \wedge v_k \in \bigwedge^k V$, a transposition of v_i and v_j introduces a minus sign

$$v_1 \wedge \cdots \wedge v_j \wedge \cdots \wedge v_i \wedge \cdots \wedge v_k = -v_1 \wedge \cdots \wedge v_i \wedge \cdots \wedge v_j \wedge \cdots \wedge v_k.$$

(4) If $\pi \in S_k$ and $v_i \in V$, then we have

$$v_{\pi(1)} \wedge v_{\pi(2)} \wedge \cdots \wedge v_{\pi(k)} = (\text{sgn} \pi) v_1 \wedge \cdots \wedge v_k.$$

Proof: (1) For $\forall w \in V$, then $w \otimes w \in I(V)$ and $w \wedge w = 0 \in \bigwedge(V)$. Thus for $\forall u, v \in V$, we have

$$0 = (u + v) \wedge (u + v) = u \wedge u + u \wedge v + v \wedge u + v \wedge v = u \wedge v + v \wedge u;$$

(2) Since both sides are linear in u and v , it suffices to prove the equation for decomposable element.

Suppose $u = u_1 \wedge u_2 \wedge \cdots \wedge u_k$ and $v = v_1 \wedge v_2 \wedge \cdots \wedge v_l$. By (1), we have $v_1 \wedge u_i = -u_i \wedge v_1$, hence

$$u_1 \wedge u_2 \wedge \cdots \wedge u_k \wedge v_1 = (-1)^k v_1 \wedge u_1 \wedge u_2 \wedge \cdots \wedge u_k$$

and therefore, we have

$$\begin{aligned} u \wedge v &= (-1)^k v_1 \wedge u_1 \wedge u_2 \wedge \cdots \wedge u_k = (-1)^k (-1)^k v_1 \wedge v_2 \wedge u_1 \wedge u_2 \wedge \cdots \wedge u_k = \cdots = \\ &= (-1)^{kl} v_1 \wedge v_2 \wedge \cdots \wedge v_l \wedge u_1 \wedge u_2 \wedge \cdots \wedge u_k = (-1)^{kl} v \wedge u; \end{aligned}$$

(3) Without loss of generality, we may assume that $i < j$ and let $a = v_{i+1} \wedge \cdots \wedge v_{j-1}$. Then it suffices to check that $v_i \wedge a \wedge v_j = -v_j \wedge a \wedge v_i$.

By (2), we have

$$(v_i \wedge a) \wedge v_j = (-1)^{\deg a + 1} v_j \wedge (v_i \wedge a) = (-1)^{\deg a + 1} (-1)^{\deg a} v_j \wedge (a \wedge v_i).$$

(4) Suppose π can be written as a product of l transpositions. Then $\text{sgn} \pi = (-1)^l$. By (3), we have $v_{\pi(1)} \wedge v_{\pi(2)} \wedge \cdots \wedge v_{\pi(k)} = (\text{sgn} \pi) v_1 \wedge \cdots \wedge v_k$. $\square$

If V is an R -module, let $V^k := V \times V \times \cdots \times V$ (k times). For the k th exterior power $\bigwedge^k V$, there is a natural map

$$\begin{aligned} \wedge : V^k &\rightarrow \bigwedge^k V \\ (v_1, v_2, \dots, v_k) &\mapsto v_1 \wedge v_2 \wedge \cdots \wedge v_k \end{aligned}$$

this map is clearly a k -linear map. By Proposition 0.2(4), it is also alternating.

Universal Mapping Property for Alternating k -Linear Maps

Theorem 0.2. : For any R -module Z and any alternating k -linear map $f : V^k \rightarrow Z$ over R , there exists a unique linear map $\tilde{f} : \bigwedge^k V \rightarrow Z$ over R such that the diagram commutes

$$\begin{array}{ccc} V^k & \xrightarrow{f} & Z \\ \downarrow \wedge & \nearrow \tilde{f} & \\ \bigwedge^k V & & \end{array}$$

i.e., $\tilde{f} \circ \wedge = f$, which means

$$\tilde{f}(v_1 \wedge v_2 \wedge \cdots \wedge v_k) = f(v_1, v_2, \dots, v_k)$$

for all $v_1, v_2, \dots, v_k \in V$.

Proof: Since $f : V^k \rightarrow Z$ is k -linear, by the universal property of tensor product, there exists a unique linear map $h : T^k V \rightarrow Z$ such that $h \circ \otimes = f$,

$$\begin{array}{ccc} V^k & \xrightarrow{f} & Z \\ \downarrow \otimes & \nearrow h & \\ T^k V & & \end{array}$$

i.e., $h(v_1 \otimes \cdots \otimes v_k) = f(v_1, \dots, v_k)$. Since f is alternating, hence

$$h(\cdots \otimes v \otimes v \otimes \cdots) = f(\dots, v, v, \dots) = 0$$

which implies that $\ker h \subseteq I^k(V)$, therefore h induces a linear map on the quotient module $\tilde{f} : T^k V / I^k V = \bigwedge^k V \rightarrow Z$ such that $\tilde{f}(v_1 \wedge \cdots \wedge v_k) = h(v_1 \otimes \cdots \otimes v_k) = f(v_1, v_2, \dots, v_k)$, i.e., we have the commutative diagram

$$\begin{array}{ccc} T^k V & \xrightarrow{h} & Z \\ \downarrow & \nearrow \tilde{f} & \\ \bigwedge^k V & & \end{array}$$

this proves the existence of $\tilde{f} : \bigwedge^k V \rightarrow Z$;

To show the uniqueness of $\tilde{f}$, note that $\tilde{f}$ is defined on all decomposable elements in $\bigwedge^k V$. $\square$

Remark 0.2. : For an R -module V , denote $L_k(V) := \text{Hom}_R(V^k, R)$ and $A_k(V) := \{f \in L_k(V) \mid f \text{ is alternating}\}$.

By the universal mapping property of tensor product, we can know that

$$L_k(V) \cong (T^k V)^\vee$$

$$f \mapsto \tilde{f}$$

where $\tilde{f} \circ \otimes = f$, i.e., $\tilde{f}(v_1 \otimes v_2 \otimes \cdots \otimes v_k) = f(v_1, v_2, \dots, v_k)$.

By the universal mapping property of the exterior power $T^k V$, we can know that

$$A_k(V) \cong \left(\bigwedge^k V\right)^\vee$$

$$f \mapsto \tilde{f}$$

where $\tilde{f} \circ \wedge = f$ i.e., $\tilde{f}(v_1 \wedge v_2 \wedge \cdots \wedge v_k) = f(v_1, v_2, \dots, v_k)$.

If V is a free R -module of finite rank, then for any k , the exterior power $\bigwedge^k V$ is also a free R -module of finite rank. Moreover, from a basis of V , we can construct a basis of $\bigwedge^k V$.

Lemma 0.2. : If $B = \{e_1, e_2, \dots, e_n\}$ is a basis for the free R -module V , then

$$e_1 \wedge e_2 \wedge \cdots \wedge e_n \neq 0.$$

It follows that for $1 \leq i_1 < i_2 < \cdots < i_k \leq n$, we have $e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_k} \neq 0$.

Proof: Define

$$f : B^n \rightarrow R$$

$$(e_{\pi(1)}, e_{\pi(2)}, \dots, e_{\pi(n)}) \mapsto \text{sgn} \pi$$

where $\pi \in S_n$ is a permutation, then f is clearly n -linear. Extend f to an n -linear map $f : V^n \rightarrow R$, which is also alternating since f is alternating on the basis elements. Indeed, $f(\dots, \sum a^i e_i, \dots, \sum b^j e_j, \dots) = \sum a^i b^j f(\dots, e_i, \dots, e_j, \dots) = -\sum a^i b^j f(\dots, e_j, \dots, e_i, \dots) = -f(\dots, \sum b^j e_j, \dots, \sum a^i e_i, \dots)$.

By the universal mapping property for alternating k -linear maps, there exists a unique linear map

$$\tilde{f} : \bigwedge^n V \rightarrow R$$

such that $\tilde{f} \circ \wedge = f$, i.e., the diagram is commutative

$$\begin{array}{ccc} V^n & \xrightarrow{f} & R \\ \wedge \downarrow & \nearrow \tilde{f} & \\ \bigwedge^n V & & \end{array}$$

particularly, we have

$$\tilde{f}(e_1 \wedge e_2 \wedge \cdots \wedge e_n) = f(e_1, e_2, \dots, e_n) = 1 \neq 0,$$

which implies that $e_1 \wedge e_2 \wedge \cdots \wedge e_n \neq 0$. $\square$

A Basis for $\bigwedge^k V$

Theorem 0.3. : If $\{e_1, e_2, \dots, e_n\}$ is a basis for V , then

$$S := \{e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_k} \mid 1 \leq i_1 < i_2 < \cdots < i_k \leq n\}$$

forms a basis for $\bigwedge^k V$.

Proof: Since $\bigwedge^k V = T^k V / I^k V$ and

$$\{e_{i_1} \otimes \cdots \otimes e_{i_k} \mid 1 \leq i_1 < i_2 < \cdots < i_k \leq n\}$$

is a basis for $T^k V$, then $\bigwedge^k V = \text{Span}(S)$;

Denote $I := (1 \leq i_1 < i_2 < \cdots < i_k \leq n)$ and

$$e_I := e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_k},$$

now suppose $\sum_I a^I e_I = 0$, where I runs over all multi-indices $1 \leq i_1 < i_2 < \cdots < i_k \leq n$ of length k . If $J = (j_1 < \cdots < j_k)$ is one particular multi-index in this sum and J' is its complement. By definition, e_J and $e_{J'}$ have no common factor, but if I and J are both of length k and $I \neq J$, then there is an e_i in e_I that is not in e_J , so that e_I and $e_{J'}$ will have the factor e_i in common. Hence we have

$$e_J \wedge e_{J'} = \pm e_1 \wedge \cdots \wedge e_n$$

$$e_I \wedge e_{J'} = 0 \text{ (if } I \neq J \text{)}$$

so we will get

$$0 = \left(\sum_I a^I e_I \right) \wedge e_{J'} = a^J e_J \wedge e_{J'} = \pm a^J e_1 \wedge \cdots \wedge e_n.$$

since $e_1 \wedge e_2 \wedge \cdots \wedge e_n \neq 0$, so we have $a^J = 0$, which means that S is linearly independent and thus S is a basis for $\bigwedge^k V$. $\square$

By this theorem, we can know that

Corollary 0.3. : If V is a free R -module of rank n , then $\bigwedge^k V$ is a free R -module of rank $\binom{n}{k}$.

Nondegenerate Pairing

Definition 0.3. : A **pairing** of two R -modules V and W is a bilinear map

$$\langle , \rangle : V \times W \rightarrow R$$

the pairing is said to be **nondegenerate** if

$$\langle v, w \rangle = 0 \text{ for all } w \in W \Rightarrow v = 0$$

and

$$\langle v, w \rangle = 0 \text{ for all } v \in V \Rightarrow w = 0.$$

Example 0.1. : An inner product on a real vector space V is a nondegenerate pairing.

If there is a pairing $\langle , \rangle$ between two R -modules V and W , then each $v \in V$ induces a linear map

$$\langle v, \rangle : W \rightarrow R$$

hence the pairing induces a map

$$V \rightarrow W^\vee$$

$$v \mapsto \langle v, \rangle$$

such map is clearly linear. Similarly, the pairing also induces an R -linear map $W \rightarrow V^\vee$ via $w \mapsto \langle , w \rangle$.

The definition of nondegeneracy says precisely that these two induced linear maps $V \rightarrow W^\vee$ and $W \rightarrow V^\vee$ are injective.

Lemma 0.3. : *Let V and W be finite-dimensional vector spaces over field R . If $\langle , \rangle : V \times W \rightarrow R$ is a nondegenerate pairing, then the induced maps $V \rightarrow W^\vee$ and $W \rightarrow V^\vee$ are isomorphisms.*

Proof: By the injectivity of the induced linear maps $V \rightarrow W^\vee$ and $W \rightarrow V^\vee$,

$$\dim V \leq \dim W^\vee \text{ and } \dim W \leq \dim V^\vee$$

by proposition 0.1, we get

$$\dim V \leq \dim W^\vee = \dim W \leq \dim V^\vee = \dim V$$

therefore, $\dim V = \dim W^\vee$ and $\dim W = \dim V^\vee$ and hence these two induced linear maps are isomorphisms. $\square$

Remark 0.3. : *In general, there is no canonical isomorphism between V and $V^\vee$, but if V is finite-dimensional and has a nondegenerate pairing, however, we can give the canonical isomorphism*

$$V \rightarrow V^\vee$$

$$v \mapsto \langle v, \rangle$$

Proposition 0.3. : *Let V be a module over a commutative ring R with identity. The multilinear map*

$$(V^\vee)^k \times V^k \rightarrow R$$

$$((\beta^1, \beta^2, \dots, \beta^k), (v_1, v_2, \dots, v_k)) \mapsto \det[\beta^i(v_j)]$$

induces a pairing

$$\bigwedge^k (V^\vee) \times \bigwedge^k V \rightarrow R.$$

Proof: For any fixed $(\beta^1, \beta^2, \dots, \beta^k) \in (V^\vee)^k$, the map

$$V^k \rightarrow R$$

$$(v_1, v_2, \dots, v_k) \mapsto \det[\beta^i(v_j)]$$

is alternating and k -linear (by the property of determinant). By the universal mapping property for $\bigwedge^k V$, there is a unique linear map $\bigwedge^k V \rightarrow R$ such that

$$v_1 \wedge v_2 \wedge \dots \wedge v_k \mapsto \det[\beta^i(v_j)].$$

So we have constructed a map

$$h : (V^\vee)^k \rightarrow (\bigwedge^k V)^\vee$$

$$(\beta^1, \beta^2, \dots, \beta^k) \mapsto h(\beta^1, \beta^2, \dots, \beta^k)$$

where $h(\beta^1, \beta^2, \dots, \beta^k)(v_1 \wedge v_2 \wedge \dots \wedge v_k) := \det[\beta^i(v_j)]$. Then h is alternating k -linear, by the universal mapping property again, there exists a unique map

$$\tilde{h} : \bigwedge^k (V^\vee) \rightarrow (\bigwedge^k V)^\vee$$

such that $\tilde{h}(\beta^1 \wedge \beta^2 \wedge \dots \wedge \beta^k)(v_1 \wedge v_2 \wedge \dots \wedge v_k) = h(\beta^1, \beta^2, \dots, \beta^k)(v_1 \wedge v_2 \wedge \dots \wedge v_k) = \det[\beta^i(v_j)]$.

Hence the map $\tilde{h}$ give rise to a bilinear map

$$\bigwedge^k (V^\vee) \times \bigwedge^k V \rightarrow R$$

$$((\beta^1 \wedge \beta^2 \wedge \dots \wedge \beta^k), (v_1 \wedge v_2 \wedge \dots \wedge v_k)) \mapsto \det[\beta^i(v_j)]. \quad \square$$

Theorem 0.4. : *If V is a free R -module of finite rank n , then for any positive integer $k \leq n$, the linear map*

$$\tilde{h} : \bigwedge^k (V^\vee) \rightarrow (\bigwedge^k V)^\vee$$

$$(\beta^1 \wedge \beta^2 \wedge \dots \wedge \beta^k) \mapsto (v_1 \wedge v_2 \wedge \dots \wedge v_k \mapsto \det[\beta^i(v_j)])$$

from the Proposition 0.3 is an isomorphism.

Proof: To simplify the notation, denote $\tilde{h}$ as f . By Corollary 0.3, we can know that both $\bigwedge^k(V^\vee)$ and $(\bigwedge^k V)^\vee$ are free R -modules of rank $\binom{n}{k}$. We prove that f maps a basis of $\bigwedge^k(V^\vee)$ to a basis of $(\bigwedge^k V)^\vee$ and therefore f is an isomorphism:

Let $\{e_1, e_2, \dots, e_n\}$ be a basis of V and $\{\alpha^1, \alpha^2, \dots, \alpha^n\}$ is the dual basis for $V^\vee$. Then

$$\{e_I := e_{i_1} \wedge \dots \wedge e_{i_k} \mid 1 \leq i_1 < i_2 < \dots < i_k \leq n\}$$

is a basis for $\bigwedge^k V$ and

$$\{\alpha^I := \alpha^{i_1} \wedge \dots \wedge \alpha^{i_k} \mid 1 \leq i_1 < i_2 < \dots < i_k \leq n\}$$

is a basis for $\bigwedge^k(V^\vee)$. Denote by e_I^* the basis for $(\bigwedge^k V)^\vee$ dual to $\{e_I\}$, i.e., $e_I^*(e_J) = \delta_{I,J}$.

We claim that $f(\alpha^I) = e_I^*$. Indeed, consider $f(\alpha^I)(e_J)$. When $I = J$, we have

$$f(\alpha^I)(e_I) = f(\alpha^{i_1} \wedge \dots \wedge \alpha^{i_k})(e_{i_1} \wedge \dots \wedge e_{i_k}) = \det[\alpha^{i_r}(e_{i_s})] = \det[\delta_s^r] = 1$$

For $I \neq J$, then some i_0 is different from $j_1, j_2, \dots, j_k$, then

$$f(\alpha^I)(e_J) = \det[\alpha^{i_r}(e_{j_s})] = 0$$

since the r_0 -th row is identically zero. So $f(\alpha^I) = e_I^*$ for any multi-indices I . Since f maps a basis to a basis, it is an isomorphism. $\square$

Remark 0.4. : *By this theorem, we can know that if V is a free R -module of finite rank, then*

$$\bigwedge^k(V^\vee) \cong (\bigwedge^k V)^\vee \cong A_k(V).$$